

Project Write-Up
WAZUH -SIEM Home Lab
Sreeram G Ramesh



1. Introduction

This project is a hands-on SIEM home lab built using Wazuh. The idea was simple: set up a small monitoring environment inside VirtualBox, connect a few virtual machines to it, and see what kind of security events get collected. The whole thing runs on a laptop using three VMs: one Ubuntu-based Wazuh manager and two monitored agents, one Linux and one Windows.

The goal was to get comfortable with how a SIEM actually works in practice, not just in theory. That means going through the full process: installation, agent enrollment, network configuration, and dashboard setup, and dealing with the errors that came up along the way.

2. Lab Architecture

The lab is made up of three virtual machines, all running inside Oracle VirtualBox. They are connected to each other using VirtualBox's NAT Network mode, which lets the VMs talk to each other while still being isolated from the outside network.

Role Details

Wazuh Manager Ubuntu Server VM, IP: 10.0.3.3

Ubuntu Agent Ubuntu VM, IP: 10.0.3.4

Windows Agent Windows 10 VM, IP: 10.0.3.5

Platform Oracle VirtualBox

Network Mode NAT Network

The Wazuh manager is the central piece. It runs the Wazuh manager service, Wazuh indexer, and the Wazuh dashboard. The two agent VMs send their security events to the manager, and everything gets visualized in the dashboard.

3. Setup Process

3.1 Wazuh Manager Setup

The first step was getting the Wazuh manager up and running on the Ubuntu server VM. The official Wazuh installer script was used for this. It installs three main things in one go: the manager (which handles the agents), the indexer (which stores the data), and the dashboard (where you can view everything).

Once installed, service status was verified using:

```
sudo systemctl status wazuh-manager
```

```
sudo systemctl status wazuh-indexer
```

```
sudo systemctl status wazuh-dashboard
```

After confirming all three services were active, the dashboard was accessible through a browser by navigating to the manager's IP address on port 443.

3.2 Ubuntu Agent Setup

A second Ubuntu VM was created and connected to the same NAT Network. The Wazuh agent package was installed on it, and the agent was

registered with the manager. The key part here is making sure the agent config file points to the correct manager IP (10.0.3.3 in this case).

After updating the config, the agent service was restarted:

sudo systemctl restart wazuh-agent

The agent then showed up in the Wazuh dashboard as active.

3.3 Windows Agent Setup

A Windows 10 VM was created and also connected to the NAT Network. The Wazuh Windows agent installer was downloaded and run. During setup, the manager IP was entered so the agent knows where to send data.

After installation, the Wazuh agent service was started from the Windows command prompt or the Wazuh Agent Manager UI. Once the service was running, the Windows agent appeared as active in the dashboard.

4. Security Events Generated

To make sure the agents were actually sending data, some test events were generated on both machines.

On the Ubuntu agent:

- Used the logger command to create custom log entries.
- Created and modified files to trigger file integrity events.
- Ran sudo commands to generate privilege escalation alerts.
- Restarted the Wazuh agent service itself, which also gets logged.

On the Windows agent:

- Used **eventcreate** in PowerShell to manually write Windows events.
- Restarted the Wazuh agent service to generate service-related logs.

All of these showed up in the Wazuh dashboard under the security events section, which confirmed that the data pipeline was working end to end.

5. Dashboard Visualisations

After the agents were sending data, a few visualizations were created in the Wazuh dashboard to make

the data easier to read. These were built using the built-in visualization tools in the Wazuh dashboard.

- **Ubuntu Agent Alerts Over Time:** a timeline chart showing when alerts were triggered
- **Ubuntu Agent Alert Severity:** a breakdown of alerts by severity level
- **Ubuntu Agent Top Alert Types:** a table showing which types of alerts came up most often

These visualizations helped turn the raw event data into something that is actually useful for monitoring, which is exactly what you would expect from a SIEM.

6. Troubleshooting

Things did not work perfectly on the first try, but that is kind of the point of a home lab. Here are the main issues that came up and how they were fixed.

Issue 1: Wazuh Installation Kept Failing

During early install attempts, the Wazuh installer would remove previously installed components after failing to complete the dashboard initialization. It kept rolling back.

What was done:

- Checked the install log at `/var/log/wazuh-install.log` to find where it was failing.
- Adjusted the VM resource allocation (more RAM, more CPU) to give it enough headroom.
- Retried the installation and verified each service step by step after it completed.

Issue 2: Unsupported Operating System Warning

The Wazuh install script threw an error saying the current Ubuntu version was not in the supported OS list.

What was done:

- Checked the Wazuh documentation for the list of supported OS versions.
- Switched to a supported version of Ubuntu before proceeding with the install.

Issue 3: Agents Showing Never Connected

After registering the Ubuntu agent, the dashboard kept showing it as Never connected. The agent was installed and running, but it was not

actually communicating with the manager.

Root cause:

The VMs were initially set up with plain NAT (not NAT Network), which means each VM gets its own isolated network and they cannot reach each other. So even though the agent was trying to connect to the manager IP, there was no actual network path between them.

What was done:

- Switched all three VMs to VirtualBox's NAT Network mode so they share the same network segment
- Updated the Wazuh agent config file with the correct manager IP (10.0.3.3)
- Restarted the Wazuh agent service after making the config changes
- Checked the agent-side log to confirm connectivity:

```
sudo tail -n 50 /var/ossec/logs/ossec.log
```

After the network fix and config update, the agent showed up as active in the dashboard.

7. Results

By the end of the project, the lab was fully working. Here is what was achieved:

- Both Ubuntu and Windows agents are enrolled and showing as active in the Wazuh dashboard
- Security events are being collected from both endpoints in real time.

- Alert visualizations are built and working in the dashboard.
- The full setup from fresh VMs to a working SIEM was completed and documented.

8. What I Learned

This lab gave a real feel for how SIEM tools work in practice. Some of the key takeaways:

- How to install and configure Wazuh from scratch on Linux.
- How agent enrollment works and what can go wrong, especially with network config.
- How VirtualBox networking modes differ: plain NAT vs NAT Network, and why it matters.
- How to read agent logs to debug connection issues.
- How to build basic dashboards and visualizations in Wazuh.
- General comfort with working across Linux and Windows endpoints in a lab environment.